

PDEs: The heat equation and its examples.

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1 Ordinary one-dimensional heat equation

Consider a thin metal rod lying along the x -axis. If one end is heated, how does the temperature spread through the rod over time? This physical process is governed by the *heat equation*

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad (1.1)$$

where $u(x, t)$ denotes the temperature at position x and time t , and $k > 0$ is the *thermal conductivity* of the material, a constant measuring how readily heat flows through it.

A fundamental property of (1.1) is that it is *linear*: the unknown u and all its partial derivatives appear only to the first power, with no nonlinear terms such as u^2 or $u \partial u / \partial x$. Linearity has an important consequence known as the *superposition principle*: if u_1 and u_2 are each solutions of (1.1), then so is any linear combination $c_1 u_1 + c_2 u_2$ for constants c_1, c_2 . This means we can construct complicated solutions simply by adding together simpler ones, a strategy we will exploit throughout. For a more mathematically rigorous explanation, you can follow [3], [2], [4], and [1]. For free and accessible material, see [Paul's Online Notes](#).

Separation of variables

To solve (1.1), we employ the technique of *separation of variables*: we hypothesise that the solution takes the product form

$$u(x, t) = X(x) T(t),$$

where X depends only on position and T depends only on time. Substituting into (1.1) and dividing both sides by $X(x)T(t)$ yields

$$\frac{T'(t)}{kT(t)} = \frac{X''(x)}{X(x)} = -\lambda, \quad (1.2)$$

where $-\lambda$ is a separation constant. We choose the constant to be negative because the boundary conditions will then produce oscillatory sine solutions, which can vanish at both endpoints.. The left-hand side of (1.2) depends only on t and the right-hand side only on x ; since these are independent variables, both sides must equal the same constant $-\lambda$. This decouples (1.1) into two ordinary differential equations (ODEs):

$$X''(x) + \lambda X(x) = 0, \quad (1.3)$$

$$T'(t) + k\lambda T(t) = 0. \quad (1.4)$$

The spatial eigenvalue problem

Equation (1.3) is the central, non-trivial one. Suppose the rod has length L and both ends are held at zero temperature, imposing the boundary conditions $X(0) = X(L) = 0$. For most values of λ , the only function satisfying both the ODE and these endpoint conditions is the trivial solution $X(x) = 0$, which carries no physical information. Only for special, discrete values of λ do non-zero solutions exist, a situation analogous to the fact that only certain frequencies (harmonics) can resonate on a guitar string fixed at both ends. These special values are called *eigenvalues* of the problem, and the corresponding non-zero solutions are called *eigenfunctions*. The problem of finding them, given a differential equation together with boundary conditions, is known as a *Sturm–Liouville eigenvalue problem*.

Solving (1.3) with the boundary conditions $X(0) = X(L) = 0$, the eigenvalues are

$$\lambda_n = \frac{n^2 \pi^2}{L^2}, \quad n = 1, 2, 3, \dots, \quad (1.5)$$

with corresponding eigenfunctions $X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$. Each eigenfunction describes one allowed “shape” of the temperature distribution along the rod.

The temporal equation

Once the eigenvalues are known from the spatial problem, the temporal equation (1.4) is straightforward to solve. It is a first-order linear ODE, and for each eigenvalue λ_n its solution is simply an exponential decay:

$$T_n(t) = e^{-k\lambda_n t} = e^{-kn^2\pi^2 t/L^2}. \quad (1.6)$$

Combining the spatial and temporal parts, a single separated solution takes the form (known as a *mode*)

$$u_n(x, t) = \sin\left(\frac{n\pi x}{L}\right) e^{-kn^2\pi^2 t/L^2}. \quad (1.7)$$

The general solution via superposition

By the superposition principle, any linear combination of modes is also a solution. The general solution is therefore the infinite sum

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-kn^2\pi^2 t/L^2}. \quad (1.8)$$

The coefficients B_n are determined by imposing the initial condition $u(x, 0) = f(x)$, which requires

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right). \quad (1.9)$$

This is a *Fourier sine series* representation of f : any sufficiently well-behaved function on $[0, L]$ can be expressed as such an infinite sum of sine waves of increasing frequency. To extract each coefficient B_n , we exploit the fact that sine functions of different frequencies satisfy (also known as orthogonality of the eigenfunctions for other details go to appendix)

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \frac{L}{2} \delta_{mn}, \quad (1.10)$$

where δ_{mn} is the Kronecker delta, equal to 1 when $m = n$ and 0 otherwise. Multiplying both sides of the Fourier series by $\sin(m\pi x/L)$, integrating from 0 to L , and using this orthogonality relation isolates each coefficient:

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (1.11)$$

General boundary temperatures

Throughout the discussion above, we assumed homogeneous Dirichlet boundary conditions

$$u(0, t) = 0, \quad u(L, t) = 0.$$

These conditions greatly simplify the analysis because they lead directly to the Fourier sine eigenfunctions. However, the same method can be extended to the more general case in which the ends of the rod are held at constant non-zero temperatures:

$$u(0, t) = T_1, \quad u(L, t) = T_2,$$

where T_1 and T_2 are constants.

In this situation, the solution no longer decays to zero. Instead, the rod approaches a non-zero steady-state temperature profile. To handle this, we first find a time-independent solution $u_s(x)$ satisfying

$$\frac{d^2 u_s}{dx^2} = 0,$$

together with the boundary conditions

$$u_s(0) = T_1, \quad u_s(L) = T_2.$$

Solving this elementary ODE gives the linear steady-state profile

$$u_s(x) = T_1 + \frac{T_2 - T_1}{L}x.$$

We now write the full solution as

$$u(x, t) = v(x, t) + u_s(x),$$

where $v(x, t)$ represents the transient part of the temperature. Substituting into the heat equation shows that $v(x, t)$ satisfies

$$\frac{\partial v}{\partial t} = k \frac{\partial^2 v}{\partial x^2},$$

with homogeneous boundary conditions

$$v(0, t) = 0, \quad v(L, t) = 0.$$

Thus the entire Fourier sine series calculation developed earlier applies directly to $v(x, t)$. The only modification is that the initial condition becomes

$$v(x, 0) = f(x) - u_s(x).$$

Hence the complete solution is

$$u(x, t) = T_1 + \frac{T_2 - T_1}{L}x + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-kn^2\pi^2 t/L^2},$$

where

$$B_n = \frac{2}{L} \int_0^L [f(x) - u_s(x)] \sin\left(\frac{n\pi x}{L}\right) dx.$$

The homogeneous case discussed previously corresponds to the special choice $T_1 = T_2 = 0$.

Long-time behaviour and convergence to steady state

The solution (1.8) reveals clearly how the temperature evolves. Each mode decays exponentially at rate $kn^2\pi^2/L^2$, which grows with n^2 , so higher modes are extinguished far more rapidly than lower ones. After a short time, the solution is dominated by the $n = 1$ (fundamental) mode, and the temperature profile settles into the shape of a single half-sine wave slowly fading to zero. The natural timescale governing this decay is

$$\tau = \frac{L^2}{k\pi^2}. \quad (1.12)$$

As $t \rightarrow \infty$, every mode vanishes and $u(x, t) \rightarrow 0$ uniformly, confirming that the rod eventually reaches zero temperature everywhere the physically expected steady state when both ends are held at zero.

Limitations of this approach

The method described above is elegant and complete for the classical one-dimensional setting, but it relies on several structural assumptions that restrict its applicability.

Linearity. The entire strategy of separation of variables depends on the equation being linear. Any nonlinear term for example a convective contribution $u \partial u / \partial x$, a reaction term u^2 , or a temperature-dependent conductivity $k(u)$ introduces cross-terms upon substitution of $u = X(x)T(t)$ that prevent the equation from splitting cleanly into separate spatial and temporal parts. The eigenvalue structure, and with it the Fourier approach, collapses entirely.

Geometry and boundary conditions. The Fourier sine basis is tied specifically to the interval $[0, L]$ with homogeneous Dirichlet conditions (temperature fixed to zero at both ends). If the boundary conditions are non-homogeneous for instance, one end is held at a fixed non-zero temperature $u(L, t) = T_0 \neq 0$ the method must first be modified. A standard fix is to subtract the steady-state solution $u_s(x) = T_0 x / L$ and instead solve for the remainder $v(x, t) = u(x, t) - u_s(x)$, which does satisfy homogeneous conditions. Only then can the Fourier sine series be applied to v .

Variable thermal conductivity. If the thermal conductivity k varies along the rod rather than being constant, the heat equation becomes

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(k(x) \frac{\partial u}{\partial x} \right), \quad (1.13)$$

which, while still linear, no longer admits the standard Fourier sine functions as its eigenfunctions. Applying separation of variables to (1.13) yields a *weighted Sturm–Liouville problem* with non-constant coefficients, whose eigenfunctions depend on the specific profile $k(x)$ and are in general not expressible in closed form. For example, if $k(x) = 1 + x$ on the rod $[0, 1]$, the spatial ODE becomes

$$\frac{d}{dx} [(1 + x) X'(x)] + \lambda X(x) = 0, \quad (1.14)$$

which is a second-order ODE with non-constant coefficients. Its solutions are not simple sines or cosines, and in general must be computed numerically. This illustrates how even a modest departure from the constant-conductivity assumption forces us beyond the classical Fourier framework.

2 A worked example: parabolic initial condition

To make the general theory concrete, we work through the heat equation on a rod of length L already heated to a temprature T with thermal conductivity k , subject to homogeneous Dirichlet boundary conditions $u(0, t) = u(L, t) = 0$ i.e the ends are then dipped in to ice or subjected to a source that maintains their boundary temprature to be 0, and the initial temperature distribution

$$f(x) = x(L - x), \quad x \in [0, L]. \quad (2.1)$$

This models a rod whose temperature peaks at the midpoint $x = L/2$ and vanishes at both ends, consistent with the boundary conditions.

Computing the Fourier coefficients

From (1.11), the coefficients are

$$B_n = \frac{2}{L} \int_0^L x(L - x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (2.2)$$

We evaluate this using integration by parts twice. Writing $I = \int_0^L (Lx - x^2) \sin(n\pi x/L) dx$ and integrating by parts:

$$I = \left[-(Lx - x^2) \frac{L}{n\pi} \cos\left(\frac{n\pi x}{L}\right) \right]_0^L + \frac{L}{n\pi} \int_0^L (L - 2x) \cos\left(\frac{n\pi x}{L}\right) dx. \quad (2.3)$$

The boundary term vanishes at both endpoints since $Lx - x^2 = 0$ at $x = 0$ and $x = L$. Integrating by parts once more:

$$\int_0^L (L - 2x) \cos\left(\frac{n\pi x}{L}\right) dx = \left[(L - 2x) \frac{L}{n\pi} \sin\left(\frac{n\pi x}{L}\right) \right]_0^L + \frac{2L}{n\pi} \int_0^L \sin\left(\frac{n\pi x}{L}\right) dx. \quad (2.4)$$

The boundary term again vanishes (since $\sin(n\pi) = \sin(0) = 0$), and the remaining integral evaluates to

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) dx = \frac{L}{n\pi} [1 - \cos(n\pi)] = \frac{L}{n\pi} [1 - (-1)^n]. \quad (2.5)$$

This is zero for even n and $2L/n\pi$ for odd n . Collecting everything, the Fourier coefficients are

$$B_n = \frac{4L^2}{n^3\pi^3} [1 - (-1)^n] = \begin{cases} \frac{8L^2}{n^3\pi^3} & n \text{ odd,} \\ 0 & n \text{ even.} \end{cases} \quad (2.6)$$

Only odd modes are excited. This is physically natural: the initial profile (2.1) is symmetric about the midpoint $x = L/2$, and only the odd-numbered sine modes share this symmetry.

The explicit solution

Substituting (2.6) into the general solution (1.8), and writing $n = 2m - 1$ to index odd integers,

$$u(x, t) = \frac{8L^2}{\pi^3} \sum_{m=1}^{\infty} \frac{1}{(2m-1)^3} \sin\left(\frac{(2m-1)\pi x}{L}\right) e^{-k(2m-1)^2\pi^2 t/L^2}. \quad (2.7)$$

Convergence to steady state

The solution (2.7) makes the convergence to steady state transparent. The m -th mode decays at rate $k(2m-1)^2\pi^2/L^2$, which grows quadratically with mode number. After one characteristic timescale $\tau = L^2/(k\pi^2)$, the fundamental mode ($m = 1$) has decayed by a factor of $e^{-1} \approx 0.368$, while the next mode ($m = 2$, corresponding to $n = 3$) has decayed by $e^{-9} \approx 0.0001$. This is already negligible. More precisely, for $t \gg \tau$, the solution is well approximated by retaining only the $m = 1$ term:

$$u(x, t) \approx \frac{8L^2}{\pi^3} \sin\left(\frac{\pi x}{L}\right) e^{-k\pi^2 t/L^2}, \quad t \gg \tau. \quad (2.8)$$

This shows that the temperature profile rapidly collapses onto the shape of a single half-sine wave, regardless of the initial condition, and then decays uniformly to zero. As $t \rightarrow \infty$, every term in (2.7) vanishes, so $u(x, t) \rightarrow 0$ pointwise for all $x \in (0, L)$. This is the physically expected steady state: with both ends held at zero temperature and no internal heat source, the rod must eventually equilibrate to zero everywhere.

The coefficients $B_n \sim n^{-3}$ decay rapidly with n , ensuring that the Fourier series converges absolutely and uniformly, and that truncating to a small number of terms already gives an excellent approximation at any fixed time $t > 0$. Here are some visualizations for this example

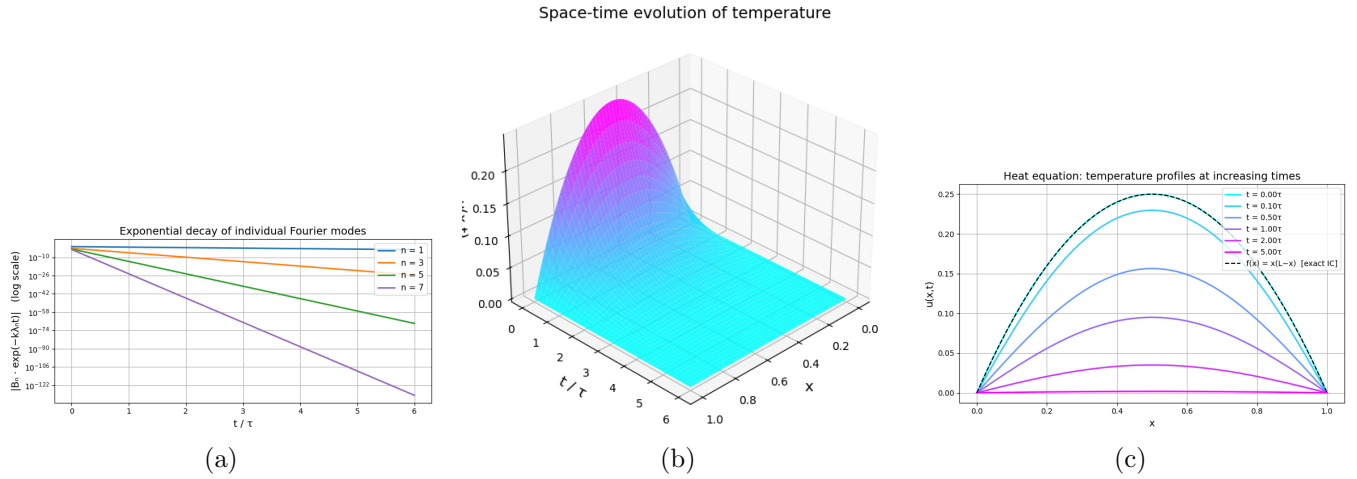


Figure 1: Simulation of the heat equation of the example discussed.

A Mathematical Toolbox

This section briefly reviews some of the mathematical ideas used throughout the discussion of the heat equation.

Ordinary and partial differential equations

An *ordinary differential equation* (ODE) involves derivatives with respect to only one variable. For example,

$$\frac{dy}{dx} = y$$

is an ODE.

A *partial differential equation* (PDE) involves partial derivatives with respect to multiple variables. The heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

is a PDE because the temperature $u(x, t)$ depends on both position x and time t .

Partial derivatives

If a function depends on several variables, we can differentiate with respect to one variable while holding the others fixed. For example, if $u(x, t)$ describes temperature, then

$$\frac{\partial u}{\partial t}$$

measures how temperature changes with time, while

$$\frac{\partial^2 u}{\partial x^2}$$

measures how curved the temperature profile is in space.

Linearity

A differential equation is called *linear* if the unknown function and its derivatives appear only to the first power and are not multiplied together.

For example,

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

is linear, while

$$\frac{\partial u}{\partial t} = u^2 + \frac{\partial^2 u}{\partial x^2}$$

is nonlinear because of the u^2 term.

Linearity is important because sums of solutions are again solutions.

Separation of variables

The method of separation of variables assumes that the solution can be written as a product

$$u(x, t) = X(x)T(t),$$

where one factor depends only on space and the other only on time. Substituting this form into the PDE allows the equation to split into two simpler ODEs.

Fourier series

A Fourier series expresses a function as an infinite sum of sine and cosine waves. For example,

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right).$$

In the heat equation, each sine function represents a different mode of temperature variation along the rod.

Orthogonality

Two functions are called *orthogonal* if their product integrates to zero over a given interval.

For sine functions,

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = 0 \quad \text{when } n \neq m.$$

Orthogonality allows us to isolate individual Fourier coefficients.

The Kronecker delta

The *Kronecker delta* is a simple notation used to express whether two indices are equal:

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$$

It is useful for writing orthogonality relations compactly. For example,

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \frac{L}{2} \delta_{mn}.$$

This single formula represents two different cases:

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = 0 \quad \text{when } m \neq n,$$

and

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2} \quad \text{when } m = n.$$

Thus, the Kronecker delta acts like an algebraic “switch” that selects only the matching mode.

Eigenvalues and eigenfunctions

An *eigenfunction* is a special function that reproduces itself (up to a constant factor) after a differential operator acts on it.

For the equation

$$X'' + \lambda X = 0,$$

the allowed values of λ are called *eigenvalues*, and the corresponding functions $X(x)$ are called *eigenfunctions*.

In the heat equation problem,

$$\lambda_n = \frac{n^2\pi^2}{L^2}, \quad X_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

Boundary and initial conditions

A PDE problem is not completely determined until additional conditions are specified.

- *Boundary conditions* describe what happens at the edges of the domain, such as

$$u(0, t) = u(L, t) = 0.$$

- *Initial conditions* specify the state of the system at the starting time, such as

$$u(x, 0) = f(x).$$

Together, these determine the unique physical solution.

Integration by parts

Integration by parts is the integration analogue of the product rule from differentiation:

$$\int u dv = uv - \int v du.$$

It is especially useful when computing Fourier coefficients involving products of polynomials and trigonometric functions.

References

- [1] Mohammed A. Al-Gwaiz. *Sturm–Liouville Theory and its Applications*. Springer Undergraduate Mathematics Series. Springer, 2008.
- [2] Richard Haberman. *Applied Partial Differential Equations with Fourier Series and Boundary Value Problems*. Pearson, 5th edition, 2012.
- [3] Jiří Lebl. *Differential equations for engineers*, 2025.
- [4] Walter A. Strauss. *Partial Differential Equations: An Introduction*. Wiley, 2nd edition, 2008.